# Chemistry

## Concise Lecture Notes

## Classification of Elements and Periodicity in Properties

### Key topics covered in the Class:

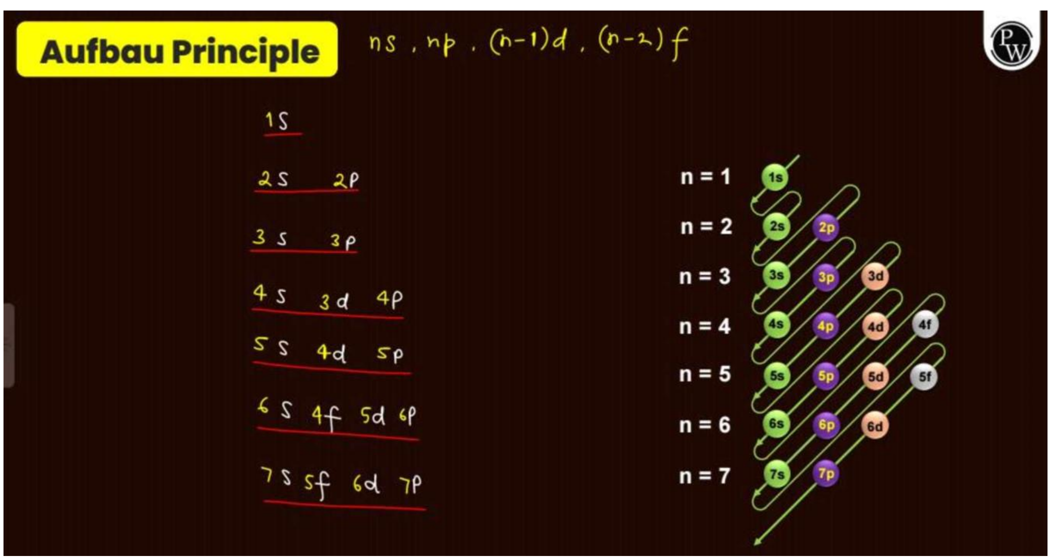
- Atom Structure, Aufbau Principle, and Electronic Configuration
- Modern Periodic Table Layout and IUPAC Naming
- Periodic Properties: Atomic Size, Ionisation Energy, Electron Affinity, and Electronegativity
- Properties of Oxides and Moseley's Equation

### Atom, Shell, Subshell

- An atom has shells for electrons. The outermost shell is the nth or *ultimate shell*, the one inside it is the (n-1)th or *pen-ultimate shell*.
- Shells are made of sub-shells (s, p, d, f), which contain orbitals. The s-subshell can hold 2 electrons, p can hold 6, d can hold 10, and f can hold 14.

### Aufbau Principle

- This principle gives the order for filling electrons into orbitals, starting from the lowest energy level.
- The filling order is: 1s, 2s, 2p, 3s, 3p, 4s, 3d, 4p, 5s, 4d, and so on.



## Modern Periodic Table Structure

- The periodic table is organised into 4 blocks (s, p, d, f), 7 periods (horizontal rows), and 18 groups (vertical columns).
- The s-block is Groups 1-2, p-block is Groups 13-18, d-block is Groups 3-12. The f-block elements (Lanthanides and Actinides) belong to Group 3.

## Element Families

- **Main Group Elements:** All elements in the s-block and p-block.
- **Alkali Metals:** Group 1 elements.
- **Alkaline Earth Metals:** Group 2 elements.
- **Transition Elements:** d-block elements.
- **Inner-transition Elements:** f-block elements (Lanthanides and Actinides).

## IUPAC name of 101 to 118

- A system is used to name newly discovered elements with atomic numbers over 100 before they get an official name.
- Names are created from roots for each digit (e.g., 1='un', 0='nil', 2='bi') and end with the suffix "-ium". For example, element 102 is Un-nil-bi-um (Unb).

## Electronic Configuration

- This describes how electrons are arranged in an atom's orbitals.
- A shorthand method uses the symbol of the nearest preceding noble gas in square brackets to represent the core electrons. For example, the configuration of Lithium (Li) is [He] $2s^1$.

## Transition Metals

- These are d-block elements whose atoms or common ions have a partially filled d-orbital ($d^1$ to $d^9$).
- Zinc (Zn), Cadmium (Cd), and Mercury (Hg) are considered d-block elements but not transition metals because they always have a completely filled d-orbital ($d^{10}$).

## Metal, Non-metal, Metalloids

• Most elements in the periodic table are metals. They are found in the s, d, and f blocks, and the lower part of the p-block.

• A few elements like Arsenic (As), Germanium (Ge), and Tellurium (Te) are metalloids, having properties between metals and non-metals.

## Effective Nuclear Charge (Zeff)

• This is the net positive charge from the nucleus that an outer electron actually feels.

• It is calculated as $Z_{effective} = Z - \sigma$, where Z is the atomic number and $\sigma$ is the shielding constant caused by inner electrons repelling the outer electron.

## Penetration Effect

• Electrons in different sub-shells get closer to the nucleus in the order: s > p > d > f.

• Because 's' electrons are closest, they feel the most attraction and are best at shielding the nucleus. 'f' electrons are the worst at shielding.

## Atomic Radius

• This is the size of an atom, measured from the nucleus to the outermost shell.

• The general order for different types of radii is: Covalent radius < Metallic radius < Van der Waals radius.

## Variation of Atomic Size in Periodic Table

• **Across a period (left to right):** Atomic size decreases because the effective nuclear charge (Zeff) increases, pulling electrons closer.

• **Down a group (top to bottom):** Atomic size increases because new electron shells are added.

## Ionic Radius

• A cation (positive ion) is always smaller than its original atom because there are fewer electrons, leading to a stronger pull from the nucleus.

• An anion (negative ion) is always larger than its original atom because the added electrons increase repulsion and weaken the nucleus's pull.

## Isoelectronic Species

- These are atoms or ions that have the same number of electrons.
- For isoelectronic species, the one with more protons (higher atomic number) will be smaller because its nucleus pulls the electrons in more tightly.

## Lanthanide Contraction

- This is the steady decrease in atomic size across the Lanthanide series (the first row of the f-block).
- It happens because the 4f electrons are very poor at shielding, causing the effective nuclear charge to increase significantly across the series.

## Effect of Lanthanide Contraction on D Block

- A key result of lanthanide contraction is that elements in the 5d series (Period 6) have nearly the same atomic size as the elements directly above them in the 4d series (Period 5).

## 3d Series (Atomic Size)

- The atomic size trend across the 3d transition series is not a simple decrease.
- It first decreases (Sc to Cr), then stays nearly constant (Mn to Ni), and finally increases slightly (Cu to Zn) due to the balance between increasing nuclear charge and increasing electron-electron repulsion.

## Ionisation Energy (IE)

- This is the minimum energy needed to remove an electron from a neutral, isolated atom in its gaseous state.
- It always takes more energy to remove subsequent electrons, so $IE_1 < IE_2 < IE_3$.

## Factors Affecting I.E.

- Higher effective nuclear charge (Zeff) and better penetration effect increase IE.
- Larger atomic size decreases IE.
- Atoms with stable electronic configurations (fully-filled or half-filled orbitals) have unusually high IE.

## How to find no of valence shell electrons?

• By looking at an element's successive ionisation energies, a large jump in energy indicates the number of valence electrons.

• For example, if the jump occurs between $IE_2$ and $IE_3$, it means the atom has 2 valence electrons.

## Electron Gain Enthalpy (öHeg)

• This is the change in energy when an electron is added to an isolated gaseous atom.

• A negative value (exothermic) means energy is released, while a positive value (endothermic) means energy is absorbed.

## Electron Affinity (EA)

• This describes an atom's tendency to accept an electron. A higher electron affinity means a stronger desire for an electron.

• Higher electron affinity corresponds to a more negative (more exothermic) electron gain enthalpy.

## Important points on Electron Gain Enthalpy

• Adding a second electron to an anion (like forming $O^{2-}$ from $O^{-}$) is always an endothermic process (positive öHeg) due to strong electron-electron repulsion.

• Elements with stable configurations like Nitrogen (half-filled p-orbital), Beryllium (filled s-orbital), and noble gases have a positive öHeg, as they resist adding an electron.

## Electron affinity of F & Cl

• An important exception to periodic trends is that Chlorine (Cl) has a higher electron affinity than Fluorine (F).

• This is because Fluorine is so small that the repulsion between its existing electrons and the new electron is very high, making the process less favourable.

## Relation between IE & EGE

- The ionisation energy of an atom is equal in magnitude to the electron affinity of its corresponding positive ion.
- For example, the energy needed to remove an electron from Na is the same as the energy released when an electron is added to $Na^+$.

## Electronegativity

- This is the ability of an atom to attract the shared pair of electrons in a chemical bond.
- It increases from left to right across a period and decreases from top to bottom down a group. Fluorine is the most electronegative element.

## Diagonal Relationship

- Some elements in the second period have similar properties to elements located diagonally below them in the third period (e.g., Li & Mg, Be & Al).
- This happens because their atomic and ionic sizes are very similar.

## Acidic Nature of Oxides

- Non-metal oxides (like $CO_2$, $SO_3$) are generally acidic, while metal oxides (like $Na_2O$, $MgO$) are generally basic.
- Some oxides (like $Al_2O_3$, $ZnO$) are amphoteric, meaning they can react as both an acid and a base.
- For a given element, the acidic nature of its oxide increases as the element's oxidation state increases (e.g., $SO_3$ is more acidic than $SO_2$).

## Modern Periodic Table (Moseley's Law)

- Henry Moseley discovered that the properties of elements depend on their atomic number (Z), not their atomic mass.
- His law states that the square root of the frequency ($\sqrt{\nu}$) of X-rays emitted by an element is directly proportional to its atomic number, as shown by the formula: $\sqrt{\nu} = a(Z - b)$.